

Deepfakes & Information Integrity

What Are Deepfakes?

Deepfakes are synthetic audio, video, or images generated by AI — typically using Generative Adversarial Networks (GANs) or diffusion models — that realistically impersonate real people. The term combines 'deep learning' and 'fake'.

Face-swap deepfakes: Replace one person's face on another's body in video with high realism.

Voice cloning: Synthesise a convincing replica of someone's voice from as little as 3 seconds of audio.

Text-to-video deepfakes: Generate entirely fabricated news segments, speeches, or events.

2025 reality: high-quality deepfakes can be created in minutes using free or low-cost tools — detection is increasingly difficult.

Risks of Deepfakes

Non-consensual intimate imagery (NCII): The most prevalent harm — 96% of online deepfakes are non-consensual sexual content targeting women.

Political disinformation: Fabricated speeches, election interference, manufactured evidence of atrocities.

Financial fraud: CEO voice cloning to authorise wire transfers — multiple \$25M+ fraud cases documented.

Reputation damage: False statements attributed to individuals, eroding trust in authentic content.

Erosion of epistemic trust: The 'liar's dividend' — even real videos can be dismissed as deepfakes.

Preventive Measures

Technical detection: AI detectors (e.g., Microsoft Video Authenticator) analyse inconsistencies in lighting, blinking, and pixel-level artefacts.

Content provenance (C2PA): Cryptographic watermarks embedded at capture time authenticate the origin of media.

Digital literacy: Teaching people to pause, verify, and cross-reference before sharing suspicious media.

Legal frameworks: Deepfake criminalisation laws are being passed in multiple jurisdictions (UK Online Safety Act, US DEFIANCE Act).

Platform policies: Social media platforms are required to label AI-generated content and remove NCII.

Key Takeaways

- Deepfakes use AI to synthesise realistic fake audio, video, and images of real people.
- Risks: NCII, political disinformation, fraud, epistemic erosion.
- Countermeasures: AI detection, C2PA provenance, digital literacy, legislation.

Quick-Revision Questions

- What technology underlies most deepfake generation?
- Explain the 'liar's dividend' — how deepfakes harm trust even in authentic content.
- What is C2PA and how does it help authenticate media?